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Problem Set #1

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No Help

"HW #1" 2/2/26

Problem #1

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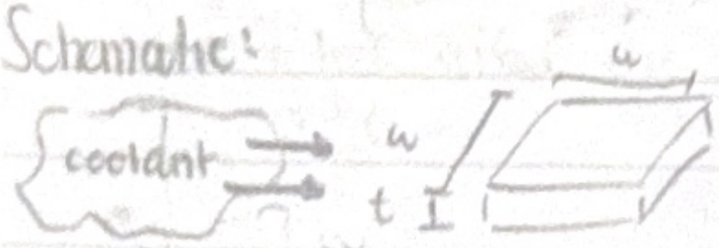
For the first part of the trip, the dominant mode of heat transfer is radiation. This is because there is no atmosphere, and in outer space, the only mode of heat transfer is radiation. For the second part, the dominant mode of heat transfer is convection. This is because there is now an atmosphere, which acts as a moving fluid.

Problem #2

Known: $k = 150\ W/mK$, $w = 5mm$, $t = 1mm$, side and back surface is thermally insulated, $T_\infty = 20°C$, $q = 9W$

Find: a) ΔT between back and front of chip @ steady state b) h_{min} if $T_{front} \leq 80°C$ (Also compare values to heat transfer coefficients in textbook)

Schematic:



Assumptions: 1) Thermal Radiation is negligible 2) Steady State

Analysis: a) $q = -kA\left(\frac{\Delta T}{t}\right)$ $A = w^2$

$$\Delta T = \frac{qt}{-kw^2} = \frac{9 \cdot 0.001}{150 \cdot 0.005^2} = \boxed{2.4°C}$$

b) $q = hA(T_s - T_\infty)$

$$h = \frac{q}{A(T_s - T_\infty)} = \frac{9}{(0.005^2)(60)} = \boxed{6000\ W/m^2K}$$

When looking at table 1.1, our value is in the range of forced convection. This means using a coolant flow with the support of a pump.

Problem #3

Known: $\varepsilon = 0.18$, $\alpha_s = 0.76$, $\sigma = 5.67 \cdot 10^{-8}\ W/m^2K^4$, $G_s = 750\ W/m^2$, $T_\infty = T_{surr} = 300K$, $h = 10\ W/m^2K$

Find: a) T_s (Can a cat travel across the roof?), b) dominant heat transfer mode (outgoing)

Schematic:

Assumptions: 1) bottom surface of roof is thermally insulated, roof is a grey body

Analysis:

a) $q'' = \varepsilon\sigma(T_s^4 - T_{surr}^4)$ $q'' = h(T_s - T_\infty)$

In: $q''_{rad} = \alpha G_s = 0.76 \cdot 750\ W/m^2 = 570\ W/m^2$

Out: $q''_{rad} = (0.18)(5.67 \cdot 10^{-8})(T_s^4 - 300^4)$

$q''_{conv} = (10)(T_s - 300)$

$570 = (1.021 \cdot 10^{-8})T_s^4 - 82.67 + 10T_s - 3000$

$3652.67 = (1.021 \cdot 10^{-8})T_s^4 + 10T_s$

$\boxed{T_s = 349.954K}$

This is too hot for the cat.

b. $\dot{q}''_{conv} = 10(T_s - 300) = 499.54 \ W/m^2$

$q''_{rad} = 0.15 \cdot 5.67 \cdot 10^{-8} (T_s^4 - 300^4) = 70.404 \ W/m^2$

The dominant mode of heat transfer is from the convection. By calculating all the flux values, it has the largest, 499.54 W/m^2.